Gaofeng Huang, Postdoc at Harvard Math Department, ghuang at math dot harvard dot edu.

Publications

Journal Articles.

J1. Huang, G. Holomorphic Approximation of Symplectic Diffeomorphisms for Calogero–Moser Spaces, *J. Reine Angew. Math.* arXiv: 2404.08505 (accepted).

J2. Andrist, R. B., Draisma, J., Freudenburg, G., Huang, G. & Kutzschebauch, F. A Criterion for the Algebraic Density Property of Affine SL_2-Manifolds, *Advances in Mathematics* **503 A.** https://doi.org/10.1016/j.aim.2026.111184 (2026).

J3. Andrist, R. & Huang, G. Direct Products for the Hamiltonian Density Property. *J. Geom Anal* **36.** https://doi.org/10.1007/s12220-025-02246-3 (2026).

J4. Huang, G. & Kutzschebauch, F. Parametric Factorization of Matrices. *J. Geom Anal* **36.** https://doi.org/10.1007/s12220-026-02328-w (2026).

J5. Huang, G., Kutzschebauch, F. & Tran, B. Unitriangular Factorization of Holomorphic Symplectic Matrices. *B. London Math. Soc.* arXiv: 2507.18963 (2026).

J6. Andrist, R. B., Freudenburg, G., Huang, G., Kutzschebauch, F. & Schott, J. A Criterion for the Density Property of Stein Manifolds, *Michigan Math. J.* **Advance Publication.** https://doi.org/10.1307/mmj/20236469 (2025).

J7. Andrist, R. B. & Huang, G. The Symplectic Density Property for Calogero–Moser Spaces. *J. London Math. Soc.* **111.** https://doi.org/10.1112/jlms.70100 (2025).

J8. Andrist, R. B., Huang, G., Kutzschebauch, F. & Schott, J. Parametric Symplectic Jet Interpolation, *Anal. Math. Phys.* **15.** https://doi.org/10.1007/s13324-025-01015-3 (2025).

J9. Huang, G., Kutzschebauch, F. & Schott, J. Factorization of Holomorphic Matrices and Kazhdan's Property (T). *Bull. Sci. Math.* **190.** https://doi.org/10.1016/j.bulsci.2023.103376 (2024).

Preprints.

PP1. Deng, F., Huang, G., Kutzschebauch, F. & Wold, E. F. *Holomorphic Approximation for Real Diffeomorphism Groups* (2026). arXiv: 2608.12565.

PP2. Huang, G. *A Counterexample to the Holomorphic Cancellation Problem* (2026). arXiv: 2610.12581.

PP3. Huang, G. & Kutzschebauch, F. *Holomorphic Approximation of Symplectic Diffeomorphism Groups via the Symplectic Density Property* (2026). arXiv: 2610.13718.

PP4. Huang, G., Kutzschebauch, F. & Rong, F. *Non-Runge Fatou-Bieberbach Domains in Stein Manifolds with the Density Property* (2026). arXiv: 2603.04903.

PP5. Huang, G., Kutzschebauch, F., Tran, N. S. & Truong, H. D. *Parametric Decompositions into Products of Involutions* (2026). arXiv: 2609.26808.

Seminar and conference presentations.

T1. *Large Holomorphic Automorphism Groups* Seminar: School of Mathematics (Zhuhai) Sun Yat-Sen University, China. Jan. 2026.

T2. *Large Holomorphic Automorphism Groups* Seminar: Geometry and Analysis Shanghai Jiao Tong University, China. Jan. 2026.

T3. *Carleman Approximation of Symplectic Diffeomorphisms for Calogero-Moser Spaces* Conference: Complex Analysis, Geometry and Dynamics IV Portoroz, Slovenia. June 2025.

T4. *Density Property and Holomorphic Dynamics* Séminare de Théorie Ergodique Jussieu Paris, France. Mar. 2025.

T5. *Large Holomorphic Automorphism Groups* Conference: KAUS Nordan Reykjavik, Iceland. May 2025.

T6. *Large Holomorphic Automorphism Groups* Seminar: Topology, Geometry and Dynamics NUS, Singapore. Jan. 2025.
T7. *Large Holomorphic Automorphism Groups* Seminar: Several Complex Variables and Complex Geometry Institute of Mathematics, CAS, Beijing. Jan. 2025.
T8. *Compatible n-tuples for the Density Property of Stein Manifolds* Seminar: Complex Analysis University of Ljubljana. Oct. 2024.
T9. *Hamiltonian Carleman Approximation of the Calogero–Moser Space* Seminar: Complex Analysis University of Ljubljana. Oct. 2024.
T10. *Hamiltonian Carleman Approximation of the Calogero–Moser Space* Conference: KAUS Nordan Östankär, Sweden. May 2024.
T11. *Large Holomorphic Automorphism Groups* Seminar: Algebra SUSTech, Shenzhen. Dec. 2024.
T12. *Symplectic Density Property of the Calogero–Moser Space* Conference: At the Confluence of Analysis, Geometry and Modern Mathematical Sciences AUB, Beirut. Oct. 2023.
T13. *Symplectic Density Property of the Calogero–Moser Space* Seminar: Groupes de Lie et espaces de modules University of Geneva. Oct. 2023.
T14. *Symplectic Density Property of the Calogero–Moser Space* Conference: Oka Theory and Complex Geometry Nordfjordeid, Norway. June 2023.
T15. *Symplectic Density Property of the Calogero–Moser Space* Seminar: Several Complex Variables and Complex Geometry Institute of Mathematics, CAS, Beijing. Apr. 2023.
T16. *Symplectic Density Property of the Calogero–Moser Space* Seminar: Symplectic Geomtry University of Neuchâtel. Mar. 2023.
T17. *Symplectic Density Property of the Calogero–Moser Space* Seminar: Symplectic Geomtry ETH Zurich. Mar. 2023.

Last updated: October 25, 2026